

HAPARANDA, Sweden

WMO: 021970

Lat: 65.824N

Lon: 24.111E

Elev: 16

StdP: 101.13

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.6	-23.9	-29.4	0.2	-26.6	-26.4	0.3	-23.8	9.5	-2.5	7.5	-3.2	1.5	0	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	24.5	17.4	22.5	16.5	20.8	15.8	19.0	22.7	17.7	21.1	16.6	19.6	2.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.6	20.6	16.3	11.6	19.1	15.3	10.9	18.1	54.0	22.6	50.0	21.3	46.7	19.6	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.4	5.5	-30.1	27.3	2.6	2.9	-32.0	29.4	-33.6	31.1	-35.0	32.7	-36.9	34.8		
(5)			-30.3	20.2	2.5	1.7	-32.1	21.4	-33.6	22.4	-35.1	23.4	-36.9	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-9.8	-9.9	-5.3	0.4	6.7	12.1	15.9	13.9	9.0	2.6	-2.8	-6.7		
	DBStd	10.10	7.18	7.25	5.07	3.25	3.90	3.19	3.04	2.95	3.35	4.30	5.40	6.59		
	HDD10.0	3261	615	556	473	287	119	14	1	5	58	231	384	518		
	HDD18.3	5887	873	789	731	537	362	189	88	139	279	489	634	776		
	CDD10.0	432	0	0	0	0	17	77	183	127	28	1	0	0		
	CDD18.3	16	0	0	0	0	1	2	11	3	0	0	0	0		
	CDH23.3	122	0	0	0	0	8	16	87	11	0	0	0	0		
	CDH26.7	19	0	0	0	0	1	3	15	1	0	0	0	0		
(14) Wind	WSAvg	2.6	2.7	2.6	2.7	2.5	2.7	2.8	2.5	2.5	2.6	2.6	2.6	2.7		
(15) Precipitation	PrecAvg	577	47	36	35	31	34	44	55	62	63	62	62	46		
	PrecMax	875	92	85	90	89	74	93	134	138	127	128	114	145		
	PrecMin	434	10	4	3	3	2	3	9	8	18	11	16	9		
	PrecStd	95	22	20	20	19	17	25	29	32	29	28	26	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	4.8	6.7	12.1	24.1	25.5	28.5	24.9	18.3	11.8	7.4	4.4		
		MCWB	2.1	2.2	2.6	6.1	14.5	16.8	19.4	18.0	13.2	10.0	6.2	3.4		
	2%	DB	1.7	2.5	4.5	9.3	19.3	22.3	25.4	22.1	16.5	10.6	6.2	3.1		
		MCWB	1.0	1.1	1.6	4.7	12.0	15.1	18.3	16.7	13.1	9.4	5.5	2.5		
	5%	DB	0.6	0.9	2.6	7.3	16.0	20.2	23.6	20.6	15.3	9.6	4.7	2.1		
		MCWB	0.2	0.4	0.7	3.6	10.1	13.8	17.4	16.2	12.8	8.5	4.1	1.6		
	10%	DB	-0.2	0.0	1.2	5.6	13.3	18.4	21.9	19.3	14.3	8.5	3.7	1.3		
		MCWB	-0.5	-0.3	0.1	2.4	8.6	13.0	16.6	15.5	12.3	7.4	3.2	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.7	3.3	7.1	15.0	18.3	21.1	19.4	15.1	10.7	6.7	3.8		
		MCDB	3.4	4.3	6.0	9.7	22.3	23.4	24.5	22.7	16.5	11.1	7.0	4.0		
	2%	WB	1.1	1.3	2.0	5.1	12.7	16.3	19.7	18.1	14.1	9.7	5.5	2.6		
		MCDB	1.6	2.1	3.7	8.8	18.2	20.4	23.6	20.7	15.5	10.4	6.0	2.9		
	5%	WB	0.4	0.5	1.1	3.8	10.8	15.0	18.6	17.0	13.4	8.6	4.3	1.8		
		MCDB	0.6	1.0	2.1	6.7	14.8	18.5	22.5	19.3	14.7	9.4	4.7	2.1		
	10%	WB	-0.6	-0.3	0.4	2.8	9.2	13.9	17.4	16.1	12.7	7.5	3.3	0.9		
		MCDB	-0.3	0.1	1.2	4.9	12.6	17.1	20.9	18.4	14.0	8.3	3.6	1.2		
(35) Mean Daily Temperature Range	MDBR		7.1	7.5	8.5	8.2	8.7	8.6	8.9	8.3	7.1	5.5	5.2	6.2		
	5% DB	MCDBR	5.7	6.0	8.1	10.9	13.3	11.3	11.8	9.9	8.1	5.7	4.5	4.3		
		MCWBR	5.6	5.3	6.1	6.9	6.8	5.3	5.7	5.3	5.5	4.6	4.4	4.2		
	5% WB	MCDBR	5.8	5.7	6.6	9.8	11.8	8.9	10.0	7.9	6.5	4.6	4.4	4.3		
		MCWBR	5.7	5.1	5.3	6.5	6.5	5.1	5.5	4.9	4.7	4.1	4.4	4.2		
(40) Clear-Sky Solar Irradiance	taub		0.206	0.272	0.302	0.337	0.345	0.348	0.365	0.349	0.325	0.301	0.343	0.355		
	taud		1.947	2.099	2.159	2.221	2.425	2.481	2.454	2.485	2.533	2.351	2.582	2.614		
	Ebn at Noon		244	580	754	821	860	864	834	813	754	565	238	74		
	Edh at Noon		19	57	87	106	96	94	94	82	63	45	18	9		
(44) All-Sky Solar Radiation	RadAvg		0.08	0.58	2.07	3.83	4.96	5.54	5.14	3.75	2.01	0.72	0.14	0.01		
	RadStd		0.01	0.11	0.20	0.27	0.45	0.42	0.45	0.33	0.23	0.13	0.02	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page